

LUMINA, AN EVENT-BASED AUTOMATED PHOTO CURATION

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ABSTRACT

This project implements Lumina, an end-to-end pipeline for organizing a personal or event photo collection. The system (1) detects people in each image with YOLOv8 [1], (2) extracts 512-dimensional appearance embeddings from person crops using torchreid (OSNet) [2], (3) clusters detections into identity groups with HDBSCAN [3], (4) computes DINO-style scene embeddings [4] and NIMA aesthetic scores [5], (5) groups images into “events” using agglomerative clustering on a feature vector that combines scene appearance with a multi-hot encoding of which person identities appear in each frame, and (6) ranks photos within each event with a seven-signal weighted score (DINO centrality, face sharpness, face size, detection confidence, pose frontality, eye aspect ratio, and NIMA). Evaluation combines internal clustering metrics (e.g. silhouette on the chosen embedding space), ablation and weight-sensitivity analyses for the scoring model, and exploratory plots (UMAP, correlation heatmaps). The Semester 1 prototype couples these models behind a FastAPI service with a React web gallery for browsing by event and identity.

Index Terms—Automated photo curation, person re-identification, event clustering, neural image assessment, web application

1. INTRODUCTION

In the era of smartphones and digital photography, users routinely accumulate hundreds or even thousands of photos from a single event—holidays, weddings, gatherings, or celebrations. Many of these images are near-duplicates captured in burst mode or repeated attempts, making it tedious and time-consuming to identify and retain only the best shots.

Lumina is an AI-powered photo management system that automatically organizes, clusters, and curates photo collections. By combining state-of-the-art computer vision models for person re-identification, scene understanding, and aesthetic evaluation, Lumina reduces redundancy while preserving meaningful, high-quality memories. The system provides an interactive web-based gallery where users can browse results organized by event and identity, with each photo scored and ranked using a multi-signal quality assessment pipeline.

1.1. Problem statement

Managing large photo collections presents three core challenges, summarized in Table 1.

Table 1. Core challenges addressed by Lumina.

Challenge	Description
Redundancy	Burst shots and repeated captures produce dozens of near-identical images, inflating storage and cluttering galleries.
Manual Effort	Reviewing, comparing, and deleting photos one-by-one is time-consuming and cognitively exhausting, especially across collections of 500+ images.
Inconsistent Criteria	Without systematic evaluation, users struggle to consistently identify the objectively “best” photo—factoring in sharpness, composition, facial expressions, and aesthetic quality simultaneously.

1.2. Project goals

The end-to-end goals are to build a system that automates photo curation by:

1. **Detecting and tracking** individuals across photos (identity clustering);
2. **Grouping photos** into meaningful events or scenes (event clustering);
3. **Scoring and ranking** images using multiple quality signals (best-image selection);
4. **Presenting results** through an intuitive, interactive interface.

2. LITERATURE REVIEW

Deep face recognition (e.g. ArcFace [6] via InsightFace [7]) supports verification and clustering from facial crops. Person re-identification extends identity cues to full-body appearance when faces are missing or unreliable [2]. Density-based clustering (HDBSCAN [3], DBSCAN) is widely used without requiring manual cluster counts. Self-supervised vision transformers in the DINO family provide strong scene embeddings [4]. NIMA [5] scores aesthetics in a learned manner.

YOLO [1] offers efficient object detection for person localization. Together, these lines of work motivate a modular pipeline; the open question for curation systems is how to fuse scene and identity cues so that event structure aligns with user expectations under limited supervision [8].

3. PROPOSED APPROACH

This section documents the Semester 1 baseline: a client-server architecture (React frontend, FastAPI backend) orchestrating the ML pipeline, together with evaluation metrics reported in the Semester 1 study. Figure 1 summarizes deployment and data flow at a high level. Figure 2 summarizes the ML pipeline

3.1. System architecture

Lumina follows a client-server design: a React [9] frontend communicates with a FastAPI [10] backend that runs the ML pipeline, packages job status, and returns structured results to the gallery UI.

3.2. Image ingestion

Users upload photos through either Google Drive (OAuth integration) or local file upload (up to 80 photos per batch). Images are downsampled to a maximum of 1280 px on the longest side to balance quality with computational efficiency.

3.3. Face detection and embedding

Model: InsightFace (buffalo_l) with 512-dimensional ArcFace embeddings [6, 7]. Each image is processed to detect faces and extract:

- **ArcFace embeddings**—512-dimensional vectors for identity matching;
- **Detection confidence scores**—reliability of the face detection;
- **Face quality signals**—sharpness (Laplacian variance), face-to-image size ratio, and pose angles (yaw, pitch, roll).

3.4. Person detection and re-identification

Models: YOLOv8 Nano for person detection + OSNet via torchreid for body re-identification [1, 2]. To handle cases where faces are occluded, turned away, or too small to detect, the system uses full-body person detection as a complementary signal:

- YOLOv8 Nano detects human bodies (class 0, input batch size 8);
- OSNet extracts body-level re-identification embeddings (256×128 input, 512-D output);
- This dual face+body approach provides robust identity tracking even when facial features are unavailable.

3.5. Identity clustering

Algorithm: HDBSCAN with `min_cluster_size=2`, `min_samples=2`, Euclidean distance [3]. HDBSCAN was selected over DBSCAN for its key advantages:

- **No epsilon parameter**—automatically determines cluster density thresholds;
- **Variable-density clusters**—handles groups of different sizes naturally;
- **Noise robustness**—outlier points are identified and reassigned to the nearest centroid rather than discarded during the pipeline process.

Clustering operates primarily on face embeddings (Euclidean distance threshold: 0.6) with OSNet body embeddings as a fallback (threshold: 0.75) for faceless detections.

3.6. Scene understanding and event clustering

Models: DINOv2 ViT-B/14 for scene embeddings + Agglomerative Clustering [4]. DINO generates scene-level embeddings that capture visual context (background, lighting, objects, spatial layout). These embeddings are used to group photos into events—for example, “ceremony at the church” versus “reception at the hotel.” Event clustering uses Agglomerative Clustering with average linkage, cosine distance metric, and a silhouette-optimized threshold sweep (range: [0.1, 3.0]) to automatically determine the optimal number of clusters.

3.7. Aesthetic scoring

Model: NIMA via pyiqa (Neural Image Assessment) [5]. Each image receives an aesthetic quality score on a 0–10 scale based on learned visual aesthetics—composition, colour harmony, exposure, and overall visual appeal. The model uses an InceptionV2 backbone trained on the AVA dataset.

3.8. Multi-signal scoring and best image selection

The core ranking system combines seven quality signals into a weighted composite score. Table 2 lists default weights (summing to 1). All signals are min–max normalized within each event group to ensure fair comparison across different lighting and scene conditions. The highest-scoring image per *event–identity* combination is selected as the “best” shot.

3.9. Frontend application

The web interface (React 19, TypeScript, Vite 6) follows a six-step flow: (1) **Landing Page**—feature overview; (2) **Photo Selection**—Google Drive (OAuth) or local upload; (3) **Processing**—real-time pipeline tracking and logs; (4) **Smart Gallery**—event-grouped results with file management; (5) **Face Analysis**—geometry breakdown and visual overlays and (6) **Ranking**—seven-signal weighted scoring per event.

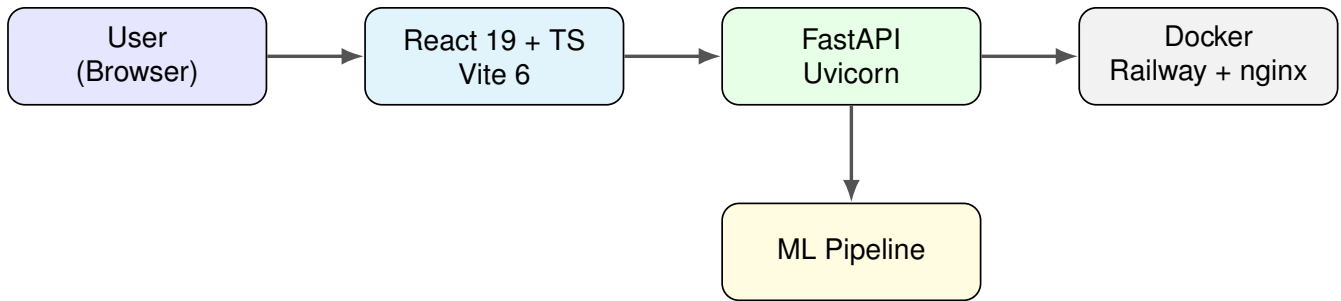


Fig. 1. Lumina tech stack: React/Vite client, FastAPI backend and deployment stack

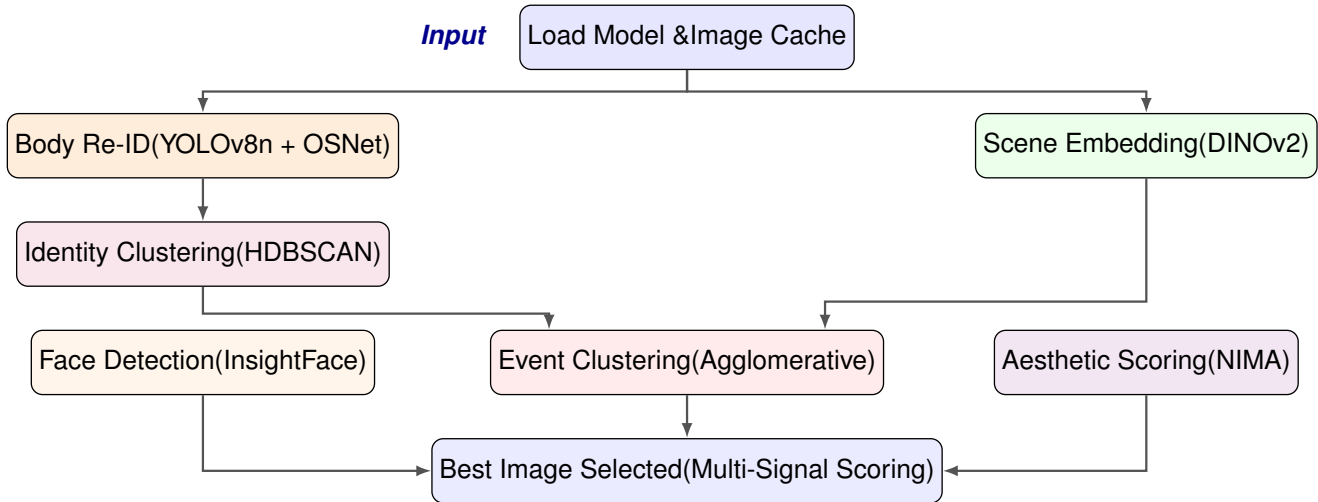


Fig. 2. Lumina ML pipeline: 5 neural networks, 2 clustering mechanism and computation to choose the best image

Table 2. Seven-signal ranker for best-image selection (Semester 1 defaults).

Signal	Weight	Description
Centrality	0.25	Cosine similarity to the event centroid (DINOv2 ViT-B/14). Measures how representative the image is of its event.
NIMA Score	0.25	Neural aesthetic quality assessment (0–10, normalized).
Face Sharpness	0.15	Laplacian variance of the face region. Higher = sharper facial details.
Face Size	0.10	Ratio of face bounding box area to total image area. Larger faces are generally preferred.
Detection Conf.	0.10	InsightFace detection score. Higher confidence correlates with clearer, more visible faces.
Pose Quality	0.10	Penalty based on yaw/pitch/roll deviation from frontal pose. Frontal-facing = highest score.
Eye Aspect Ratio	0.05	Measures whether eyes are open. Prevents selection of photos with closed or blinking eyes.
Total	1.00	

3.10. Technology stack

Table 3 lists principal technologies used in the Semester 1 implementation.

Table 3. Technology stack (Semester 1).

Layer	Technologies
Frontend	React 19.2, TypeScript 5.8, Vite 6.2, TailwindCSS, Lucide React
Backend	FastAPI, Uvicorn, Python 3.x
ML/CV	PyTorch, Transformers (DINOv2), InsightFace, torchreid (OSNet), Ultralytics (YOLOv8), pyiqa (NIMA), MediaPipe, OpenCV
Clustering	scikit-learn (HDBSCAN, AgglomerativeClustering) [8]
Deployment	Docker, Railway, nginx (reverse proxy)

3.11. API design

Table 4 summarizes the core REST endpoints exposed by the FastAPI backend.

Table 4. Representative API endpoints.

Endpoint	M.	Purpose
/api/health	GET	Health check
/api/analyze	POST	Submit batch (returns job ID)
/api/analyze/{jId}	GET	Poll progress & results
/api/face-analysis	POST	Submit face geometry job
/api/face-analysis/{jId}	GET	Poll face analysis results

4. EVALUATION AND ANALYSIS

This section reports the clustering and ranking metrics from the Semester 1 evaluation. Table 5 summarizes the four primary metrics used.

Table 5. Evaluation metrics used in this study.

Metric	Interpretation
Silhouette	Compares within-cluster cohesion to separation; ranges $[-1, 1]$, higher is better.
Calinski–Harabasz	Ratio of between-cluster to within-cluster dispersion; higher indicates more separated, compact clusters.
Davies–Bouldin	Average similarity between each cluster and its nearest neighbour from centroids; lower is better.
Kendall’s τ	Rank correlation between full and ablated orderings; 1 = identical, 0 = no association.

4.1. Re-ID clustering performance

Table 6 compares HDBSCAN (primary method) with the best DBSCAN configuration found via an ε -sweep over 21 values (18 yielding valid metrics).

Table 6. Re-ID clustering metrics: HDBSCAN vs. DBSCAN-opt ($\varepsilon = 0.20$).

Metric	HDBSCAN	DBSCAN-opt
Silhouette	≈ 0.254	$\approx$ 0.787
Calinski–Harabasz	≈ 9.2786	$\approx$ 55.6
Davies–Bouldin	≈ 1.2555	$\approx$ 0.224

4.2. Event clustering (DINO + person multi-hot)

Event clustering uses L2-normalized features built from DINO image embeddings and a person co-attendance multi-hot block, with cosine distance for agglomerative clustering. Silhouette score is the primary metric for selecting linkage distance threshold and `event_agg_person.weight` (grid 1.0–5.0,

step 0.5). Over 16 images with 18 person columns, best silhouette increased monotonically with person weight: from ≈ 0.764 at weight 1.0 (5 clusters) to ≈ 0.775 at weight 5.0 (threshold 0.35, 7 clusters). Table 7 reports the full tuning results. The notebook selects weight 5.0, yielding 7 event groups—i.e., a stronger co-attendance signal modestly improves silhouette and splits events further on this corpus.

Table 7. Person-weight tuning: best silhouette over distance threshold sweep.

w_{person}	Silhouette	Threshold	Clusters
1.0	0.7644	0.60	5
1.5	0.7674	0.35	7
2.0	0.7708	0.35	7
2.5	0.7725	0.35	7
3.0	0.7735	0.35	7
3.5	0.7741	0.35	7
4.0	0.7745	0.35	7
4.5	0.7748	0.35	7
5.0	0.7750	0.35	7

4.3. Ablation study

Table 8 summarizes event-level ranking stability when one signal is removed and remaining weights are renormalized. Only events with at least two photos are included (7 qualifying events).

Table 8. Ablation: Kendall τ vs. full model and top-1 changes.

Removed Signal	Kendall τ	Top-1 Δ
Eye Aspect Ratio	1.00	0
Detection Score	≈ 0.78	2
Centrality	≈ 0.67	1
NIMA	≈ 0.56	1
Pose Quality	≈ 0.44	2
Face Size	≈ 0.33	2
Sharpness	≈ 0.33	3

Key observations: EAR is perfectly stable ($\tau = 1.0$, zero top-1 changes), suggesting it rarely disambiguates among top candidates. Sharpness and face size are the most disruptive when removed ($\tau \approx 0.33$, 3 and 2 flips respectively), indicating these signals strongly influence final rankings. Centrality and NIMA sit mid-range ($\tau \approx 0.67$ and ≈ 0.56 , one flip each). With only 7 qualifying events, 3/7 flips for sharpness should be read as small- N sensitivity rather than population-level robustness.

4.4. Weight sensitivity ($\pm 20\%$ perturbation)

For each base weight, scores are recomputed with that weight multiplied by 0.8 or 1.2 *without* renormalizing the full weight

vector; the number of events whose top-1 image changes is counted. All signals except NIMA produced zero top-1 flips under $\pm 20\%$ perturbation. NIMA at $+20\%$ caused one flip (none at -20%), yielding a total sensitivity count of 1. This demonstrates that the default weights are empirically stable on this dataset: the winner is almost insensitive to small weight jitter, except for a marginal case when up-weighting NIMA.

4.5. UMAP visualization

Two-dimensional UMAP projections of ArcFace and DINO embeddings serve as qualitative structure checks. UMAP is a non-linear dimensionality reduction technique; distances are not strictly meaningful globally, so clusters and overlap should be interpreted qualitatively only.

5. ITERATION AND EXPLORATION

This section documents the iterative development process and key insights gained during experimentation.

5.1. OSNet person Re-ID improves silhouette score

The first attempt at event clustering used only DINOv2 global scene embeddings. While the results showed decent qualitative structure, introducing person Re-ID features from OSNet significantly improved the silhouette score. Tables 9 and 10 compare the threshold sweep results with and without person Re-ID (evaluated on separated test).

Table 9. Threshold sweep *with* person Re-ID (10 images, DINO 768-D + person 10-D, $w_{\text{person}} = 1.0$).

Threshold	Clusters	Silhouette
0.10	5	0.8398
0.35	4	0.9418
0.60	4	0.9418
0.85	3	0.7149
1.10–2.85	1	NaN

Table 10. Threshold sweep *without* person Re-ID (DINO scene embeddings only).

Threshold	Clusters	Silhouette
0.10	5	0.7495
0.35	4	0.9033
0.60	3	0.7991
0.85	2	0.5185
1.10–2.85	1	NaN

Adding person Re-ID features improved the best silhouette from 0.9033 to 0.9418—an improvement of approximately 4.3%. The combined features also produce more stable cluster counts at optimal thresholds.

5.2. Managing the weight of person Re-ID against DINO

On the next iteration, the dataset was expanded and deliberately included two separate occasions taken from the same house (two separate gatherings). Originally, the model did not manage to separate these clusters. The likely reason is that the DINO embedding dimensionality (768-D) is much larger than the person cluster multi-hot block, causing scene similarity to dominate. To address this, a person-weight multiplier w_{person} was introduced, which multiplies the weight of the person cluster embedding segment. The value is automatically selected via a silhouette sweep over the grid $[1.0, 5.0]$ at step 0.5. As shown in Table 7, increasing w_{person} to 5.0 enabled the model to successfully split the two gatherings that were originally merged, yielding 7 distinct event clusters with a best silhouette of 0.7750.

6. RESULTS SUMMARY

Table 11 consolidates key results from the Semester 1 evaluation.

Table 11. Consolidated Semester 1 results.

Result	Value / Finding
Re-ID Silh. (HDB-SCAN)	≈ 0.254
Re-ID Silh. (DBSCAN-opt)	≈ 0.787
Calinski–Harabasz	≈ 55.6
Davies–Bouldin	≈ 0.224
Event Silh. ($w = 5.0$)	0.7750 (7 clusters)
Weight Sensitivity	1 flip total (NIMA +20%)
Most Disruptive Ablation	Sharpness (3/7 top-1 changes)
Most Stable Ablation	EAR ($\tau = 1.0$, 0 changes)

The incremental improvement in silhouette scores across iterations confirms the value of combining person co-attendance signals with scene embeddings. The multi-signal ranker demonstrates robustness under weight perturbation, with sharpness and face size identified as the most influential individual signals for final image selection.

7. DISCUSSIONS AND LIMITATIONS

The current system has several areas for improvement:

- **Fixed scoring weights**—the system does not adapt to individual user preferences, since there is no feedback loop to help it learn what each person values in a photo.
- **Full recomputation**—when new photos are added, the entire collection must be reprocessed from scratch, as there is no caching or persistence of embeddings.
- **Limited explainability**—selections are made in a “black box” manner, with no explanations provided for why certain

images are chosen.

- **No semantic search**—users cannot search by scene description or visual similarity.
- **Rigid selection policy**—the curation strategy always selects just one best image per group, without options for more diverse or configurable results.
- **Single clustering strategy**—for identity clustering, the system relies on a single method and does not fuse face and body detection for greater accuracy.

UMAP and clustering metrics complement, but do not replace, human judgment of album quality.

8. CONCLUSIONS AND FUTURE WORK

Lumina successfully addresses the problem of large photo collections by providing an automated, end-to-end curation pipeline that mitigates redundancy, reduces manual effort, and applies consistent quality criteria. The system integrates multiple state-of-the-art computer vision models—including OSNet for re-identification and DINOv2 for scene understanding—to cluster photos by identity and event. Evaluation confirmed that the multi-signal ranker is robust and effective, with key metrics like Re-ID silhouette scores (up to ≈ 0.787) and stable top-1 selections under weight perturbation. Ultimately, Lumina meets its goal of delivering a functional and explainable prototype for intelligent photo management, drastically simplifying the user experience for reviewing and selecting high-quality memories.

8.1. Future directions

To address the limitations described in Section 7, several extensions are planned:

- **User-specific fine-tuning**—fine-tuning the core vision models on each user’s personal photo collection to better understand unique styles.
- **Federated preference learning**—learning from many users to improve personalization without sharing raw photos, protecting privacy.
- **Video key-frame extraction**—expanding the platform to handle video content.
- **On-device and offline modes**—running efficiently on mobile devices, even offline.
- **Richer event signals**—using additional data like GPS, timestamps, audio, and calendar information to improve event grouping.
- **Active learning**—the system will ask users for help with uncertain cases, making clustering more accurate and efficient.

9. AUTHOR CONTRIBUTIONS

Qing Quan was responsible for the backend model pipeline integration, web application development, and deployment of

the model and application. Benyamin was responsible for the Python notebook design of the model and report writing.

10. AI TOOL DECLARATION

Cursor was used as the primary AI tool for code writing and to generate the initial structure of the report. The authors are responsible for the content and quality of the submitted work.

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